

GIANLUIGI VITALE

gianluigi.vitale12@gmail.com | Pistoia, Italy

EDUCATION

B.Sc. in Computer Engineering — *Universitas Mercatorum, Rome, Italy* *Expected Feb. 2027*

GPA: 29.03/30 (Italian scale) · 3.79/4.0 (WES-converted)

Thesis: DSPProfiler - Static Analysis and Machine Learning for Automated Detection of Data Structure Performance Anti-patterns in Python

Supervisor: Prof. Rocco Pietrini

Scientific High School Diploma — *Liceo Scientifico "S. Cannizzaro," Palermo* *2014–2019*

Full Stack Web Development Bootcamp — *Boolean SRL (700 hours)* *Jan.–Jun. 2020*

PUBLICATIONS

G. Vitale, "DriftBench: Measuring and Predicting Infrastructure Drift in LLM Serving Systems," *Proceedings of MLSys 2026*, Bellevue, WA, May 2026. Oral presentation. DOI: 10.5281/zenodo.19361066. Recorded talk: <https://youtu.be/KnsVj5DdlEo>

UNDER REVIEW

G. Vitale, "Decomposing One Professional-Framing Pipeline: Which Components Shift LLM Safety Boundaries?," *NeurIPS 2026 main track*. Submitted to NeurIPS 2026. First-author manuscript.

G. Vitale, "Natural Language Is Flatland for Machine Intelligence: AI Systems Should Decouple Their Reasoning Substrate from Human Language," *NeurIPS 2026 position paper track*. Submitted to NeurIPS 2026. First-author manuscript.

RESEARCH INTERESTS

Machine learning systems, LLM serving infrastructure, accelerator portability, TPU kernels and numerical behavior, ML benchmarking, inference reliability, and LLM safety evaluation.

PROFESSIONAL EXPERIENCE

Systems Engineer — *Italian Ministry of Culture* *Apr. 2024–Present*

Designed and deployed a Retrieval-Augmented Generation system over 10,000 internal administrative documents, delivering source-cited answers to staff queries while minimizing hallucinations. Pioneered the first AI-assisted public consultation interface in an Italian state archive, enabling both employees and citizens to search and cross-reference the full inventory database through natural language, fundamentally transforming how the institution serves its users.

Web Developer — *Informamuse S.R.L.* *Oct. 2020–Apr. 2021*

Developed websites and web applications using HTML, CSS, JavaScript, PHP, Laravel, Vue.js, and MySQL. Managed projects end-to-end from design through deployment.

ACADEMIC SERVICE

Ethics Reviewer — *NeurIPS 2026* *2026*

Reviewed submissions for ethical concerns, potential societal harms, research-integrity risks, and compliance with the conference's ethics guidelines.

Artifact Evaluation Committee Member — SOSP 2026

2026

Evaluated artifacts for SANDHI, which adaptively merges model components to reduce the memory footprint of co-located fine-tuned models, and M2K, which makes the model-kernel interface explicit to detect memory bugs in CUDA kernels used by LLM inference systems.

Artifact Evaluation Committee Member — MLSys 2026

2026

Reviewed four accepted papers covering attention kernel optimization (ThunderKittens/FlashAttention-3), full-stack ML profiling (XProf), GPU memory disaggregation, and distributed inference. Reproduced experimental results on rented GPU clusters (8×H100 via RunPod) and AMD HPC infrastructure (MI350X through the AMD University Program). Recommended the artifact that ultimately received the Distinguished Artifact Award.

RESEARCH EXPERIENCE

- **DriftBench and TPU validation:** Built a measurement and prediction framework spanning 236,985 prompt-response pairs across 105 configurations. The Portability Risk Index generalizes to unseen hardware ($R^2 = 0.909$). Extended the result to TPU v5e/v6e, where predictability held across a new numerical path ($R^2 = 0.72$ - 0.96).
- **DeepSeek V4-Flash on TPU:** Completed a 32-chip v4-64 bring-up with every attention mode validated and both prefill and decode operational. Equivalent support is now implemented in vLLM TPU inference through parallel ecosystem work, independently of my drafted PRs.
- **DSA kernel for TPU:** Developed a working implementation that is bit-exact against the reference through 8K tokens and runs through 256K. Work in progress; it is the foundation of the GLM-5.2 port.
- **Bit-level TPU MXU characterization:** Reverse-engineered TPU v5e/v6e numerical behavior and built a CPU simulator that reproduces real Llama-3.1-8B matrix multiplications bit-for-bit on a native pass, enabling deterministic study of TPU arithmetic.
- **Long-context genomics with CNR Italy:** Collaborating on whether long-context DNA models encode linkage disequilibrium, combining ML systems work with genomics expertise at Italy's National Research Council.
- **Structured three-stage framing:** Conducted approximately 31,900 trials across nine LLMs using a factorial ablation of a professional-framing jailbreak, isolating the effects of persona framing, terminology substitution, and moral justification.

TECHNICAL SKILLS

Languages: Python, JavaScript, PHP, SQL, HTML/CSS, Bash

Frameworks: PyTorch, JAX, Pallas, Hugging Face Transformers

ML/Systems: vLLM, tpu-inference, SGLang, TensorRT-LLM, CUDA, Docker

Infrastructure: Google Cloud TPU (v4, v5e, v6e), Linux administration, networking, cloud GPU provisioning (RunPod, AMD HPC/Chameleon Cloud), CI/CD

Research Tools: LaTeX, Git, Jupyter, scipy, statsmodels

COMPUTATIONAL RESOURCES

- Google Cloud TPU Research Cloud (TRC) - active since May 2026. Current 320-chip allocation: 32 on-demand v4 plus 288 spot chips across v4, v5e, and v6e in US and EU zones.
- AMD University Program account (active since March 2026)
- Chameleon Cloud allocation (2,000 SUs)
- RunPod GPU cluster experience (H100, H200, B200, MI300X configurations)

LANGUAGES

Italian (native), **English** (C1 — academic proficiency, scientific publications)

CONTACTS & PROFILES

Google Scholar: <https://scholar.google.com/citations?hl=en&user=x4K6NWUAAAAJ>

ORCID: <https://orcid.org/0009-0007-6740-2695>

OpenReview: https://openreview.net/profile?id=~Gianluigi_Vitale1

Portfolio: <https://gianluigivitale.github.io/portfolio/>

GitHub: <https://github.com/GianluigiVitale>

LinkedIn: <https://www.linkedin.com/in/gianluigi-vitale615>